

ProLiVis 2.0: literature-centric visualization of protein–protein interaction networks, with a citation-trust model for interaction evidence

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Abstract

Protein–protein interaction databases record evidence without weighing it. In BioGRID, an interaction asserted once by a single high-throughput screen and one confirmed by twenty laboratories across a dozen assays are the same kind of row in the same file. Tools built on such databases inherit that flattening: they draw every reported interaction as an edge, and the resulting picture states that two proteins interact without stating how much anyone should believe it.

We present ProLiVis 2.0, a rewrite of the literature-centric visualization system of [1]. It contributes three things. First, a *citation-trust model* that scores each interaction from seven terms, including a term for the number of *independent laboratories* behind the supporting publications, obtained by clustering those publications over shared institutional affiliations; a plain count of publications cannot distinguish five confirmations from one group publishing five times. Second, a deterministic reformulation of the center layout, closed-form and $O(n \log n)$, which replaces the force-directed placement of the original and makes published figures regenerable from a session manifest. Third, an implementation that runs entirely in a web browser, with an embedded analytical database, requiring no installation and uploading no data.

On BioGRID release 5.0.260 restricted to SARS-CoV-2, 24,344 of 34,540 reported interactions — 70% — rest on a single publication, and raising the trust threshold to 0.2 leaves 11,320 of them. That the large majority of a curated interaction network is unreplicated is a fact no existing view of the database makes visible.

1 Introduction

BioGRID [2] curates over two million reported protein and genetic interactions. Each record states that one publication, using one experimental system, reported that two genes interact. It does not state whether the assay was appropriate, whether anyone reproduced the result, or whether the reporting publication was ever cited. Those facts exist in the records — distributed across many rows — but no field holds them, and no standard view surfaces them.

The consequence is familiar to anyone who has drawn such a network. Every reported interaction becomes an edge; edges are indistinguishable; and the picture asserts uniform confidence in claims whose evidential standing differs by orders of magnitude. Filtering by experimental system or by throughput helps, but both are per-record properties, and the question “how well supported is *this interaction*” is a property of the set of records supporting it.

ProLiVis 1.0 [1] proposed treating the literature as the primary object of visualization rather than as metadata: publications and experimental methods become nodes in their own right, arranged in a three-level hierarchy around an organism. The idea addressed the right problem. Its implementation had three limitations that this paper removes.

1. Publications were attached to a single, arbitrarily chosen experimental method. The grouping query selected a non-aggregated column, so a publication reporting three assays was filed under whichever the query planner returned first.

2. Layout was delegated to a separate force-directed executable. Identical input produced different pictures on different runs, and the executable is no longer available, so the published figures cannot be regenerated.
3. Evidence was displayed but never weighed. The tool could show that a publication reported an interaction; it could not say what that was worth.

1.1 Contributions

- A seven-term citation-trust model (§3) combining evidence internal to BioGRID with external bibliometric data. Its distinguishing term estimates the number of independent research groups behind an interaction by clustering the supporting publications over shared institutional identifiers.
- A deterministic center layout (§4) in which angular position encodes experimental method and radial position encodes hierarchy level, with closed-form placement and no iterative relaxation.
- Structural extraction under an evidence threshold (§5), including edge removal in ascending order of trust, which asks what network structure survives if only well-supported interactions are believed.
- A recursive high-level reading of a network too large to draw (§5): modules as nodes and evidence as links, with any module openable into its own high-level graph. The grouping is chosen per level, because the structural decomposition that works on a sparsely studied organism returns a densely studied core unchanged.
- A zero-installation implementation (§6) that ingests full BioGRID releases in the browser, keeps user data local, and emits session manifests sufficient to reproduce any figure.

2 Related work

Network visualization. Cytoscape [3] is the standard platform for biological network analysis; Gephi and Graphia serve general graph visualization. All take interactions as given and provide no vocabulary for evidence quality beyond whatever attributes the input file carries. The earlier PPI-specific tools surveyed by [1] — ProViz, Osprey, Medusa, PIVOT, Robinviz — share this property.

Confidence scoring. STRING [4] assigns confidence scores by combining evidence channels, but its experimental channel is a benchmarked probability derived from co-complex association, not an account of how many independent groups reported a given interaction. The IMEx consortium’s MIscore [5] scores interactions from number of publications, interaction detection method and interaction type — the closest existing work to ours, and the natural comparison. MIscore’s publication term is a count; ours is a count of *groups*, and we add terms for bibliometric impact and recency. We regard MIscore as a baseline rather than a competitor and report against it in §7.

Literature-centric views. Making publications first-class nodes is unusual. The nearest neighbours are citation-network tools, which visualize papers without the interactions they report, and provenance-aware curation interfaces, which record evidence without drawing it.

3 The citation-trust model

Let e be an undirected protein pair. Write $R(e)$ for the BioGRID records supporting it, $P(e)$ for the distinct publications among them, and $S(e)$ for the distinct experimental systems. Each term below maps to $[0, 1]$, or to $\perp$ (unknown) when its input is unavailable.

3.1 Terms

Replication.

$$t_{\text{rep}}(e) = 1 - \exp\left(-\frac{|P(e)| - 1}{\kappa_p}\right), \quad \kappa_p = 1.5. \quad (1)$$

Zero at $|P| = 1$: a single report is a claim. Saturating, so the second publication contributes far more than the tenth.

Independence. Let $\sim$ be the relation on $P(e)$ holding between two publications that share at least one institutional identifier, and let $L(e)$ be the set of equivalence classes of its transitive closure — computed by union-find over ROR identifiers obtained from OpenAlex [6], falling back to the first-author label where affiliations are unresolved. Then

$$t_{\text{ind}}(e) = 1 - \exp\left(-\frac{|L(e)| - 1}{\kappa_l}\right), \quad \kappa_l = 1.2. \quad (2)$$

This is the term a publication count cannot express. Consider two interactions each supported by five publications: one by five separate laboratories, the other by one laboratory five times. Their replication terms are identical by construction; their independence terms are $t_{\text{ind}} \approx 0.96$ and $t_{\text{ind}} = 0$. The term returns $\perp$ when no provenance is resolvable, rather than asserting that one publication implies one group.

Method diversity. Assays are partitioned into evidence classes — binary, co-complex, structural, proximity, enzymatic, colocalization, genetic — grouped by shared failure mode. Writing $C(e)$ for the classes spanned,

$$t_{\text{div}}(e) = 1 - \exp\left(-\frac{(|C(e)| - 1) + \gamma(|S(e)| - |C(e)|)}{\kappa_d}\right), \quad \gamma = 0.3, \kappa_d = 1. \quad (3)$$

Two affinity-capture experiments can be misled by the same sticky bait; an assay from another class fails differently. Repeating within a class therefore earns only partial credit γ relative to spanning classes.

Method directness. $t_{\text{dir}}(e) = \max_{s \in S(e)} d(s)$, where d is a per-assay prior on how directly the assay demonstrates physical contact, ranging from 1.0 for co-crystal structure to 0.2 for co-localization and 0 for every genetic system. Assays absent from the bundled vocabulary receive 0.4, so a newly introduced high-throughput method cannot inflate scores before classification.

Throughput. The low-throughput share of $R(e)$; $\perp$ when untagged.

Literature impact. For publication p with c_p citations and age a_p years, let $r_p = \log(1 + c_p/a_p)$. Then $t_{\text{imp}}(e)$ is the empirical percentile of $\max_{p \in P(e)} r_p$ within the dataset’s corpus of citation rates. A rate rather than a count, so recent work is not penalised for its age; a percentile within the corpus, because absolute citation counts are not comparable across fields; the maximum rather than the mean, so one landmark paper is not diluted by routine papers that also mention the interaction. Unresolved publications yield $\perp$, never 0: scoring our own ignorance as absence of impact would systematically penalise older and less-indexed literature.

Currency. $t_{\text{cur}}(e) = 2^{-\Delta/25}$ where Δ is years since the most recent supporting publication.

3.2 Combination

Given weights w_i , the score is a weighted mean over the *informed* terms, with the weights of unknown terms redistributed:

$$\text{trust}(e) = \frac{\sum_{i:t_i \neq \perp} w_i t_i}{\sum_{i:t_i \neq \perp} w_i}, \quad \text{coverage}(e) = \frac{\sum_{i:t_i \neq \perp} w_i}{\sum_i w_i}. \quad (4)$$

Treating $\perp$ as a mid-range value would let missing data pull every score toward the mean and make an unenriched dataset indistinguishable from a genuinely ambiguous one. Coverage is reported alongside every score and included in every export: a score of 0.6 at 40% coverage is a different claim from the same score at full coverage.

Three weight presets are shipped: **literature-aware** (default), **evidence-only** (offline; bibliometric terms zeroed), and **structural-strict** (directness-dominated). Weights are a documented default, not a fitted optimum, and are editable.

4 Center Layout 2.0

The layout places three levels of node on concentric rings: the organism at the origin, experimental systems on the first ring, publications on an outer band (Figure 1).

Sector allocation. Each system s receives an angular sector of width proportional to $\max(\pi_s, \phi)$, where π_s is its share of publications and $\phi = 0.022$ is a floor ensuring that a method used by a single paper remains a selectable target; widths are renormalized after flooring.

Publication placement. A publication using a single method is assigned to that method’s sector. A publication using several is assigned to the sector nearest the circular mean of its methods, computed by averaging unit vectors — arithmetic mean of angles is wrong on a circle — and receives an edge to *every* method it used. This is the correction to the defect described in §1: one publication is one node, and its methods are all represented.

Within a sector, publications are packed into rows filling the band, with the capacity of row k at radius ρ_k given by $\lfloor \theta_s \rho_k / \sigma \rfloor$ for sector width θ_s and slot size σ . Row spacing shrinks so that the outermost row remains inside a fixed band. Both properties are necessary at real scale: letting rows march outward without bound causes a method with a thousand publications to push its fan to several times the method ring’s radius, collapsing every other method into an unreadable sliver.

Determinism. Inputs are sorted canonically, ties break on identifier, and no random source or clock is consulted; string comparison is explicitly locale-independent, since **localeCompare** would otherwise permit two machines to order method names differently. Identical input therefore yields byte-identical coordinates. Complexity is $O(n \log n)$, dominated by the sort.

5 Structural extraction under an evidence threshold

Trust scores are edge weights, so every standard decomposition becomes available as a function of how much evidence one demands: maximal cliques by Bron–Kerbosch with pivoting over a degeneracy ordering, biconnected components and articulation points by Hopcroft–Tarjan, k -cores, and contraction to a high-level graph whose inter-module edges carry the summed trust of the original edges spanning them.

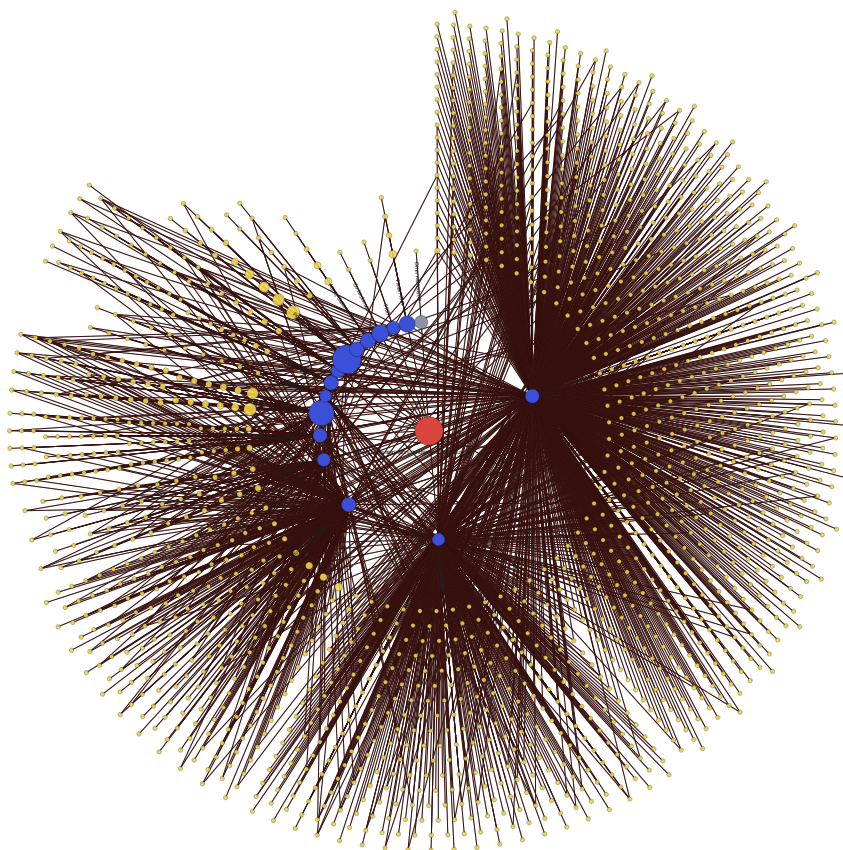


Figure 1: Center Layout 2.0 for SARS-CoV-2 (BioGRID 5.0.260): 1,688 publications across 19 experimental methods, with methods supported by fewer than five publications folded into a single node. Sector width encodes share of the literature; node area encodes interactions contributed. The two encodings disagree, and informatively so: the widest sectors (Reconstituted Complex, Co-crystal Structure) are where most *papers* are, while the largest nodes (Affinity Capture-MS, Proximity Label-MS) are where most *interactions* are.

5.1 Reading a network one level up

A network of 10^5 interactions cannot be read as a node-link diagram at any layout quality. Contraction answers this only if it can be applied repeatedly: the modules of a large network are themselves large. We therefore define a drill-down in which a high-level node expands into a high-level graph of its own, and iterate until a level is small enough to draw as proteins.

Which grouping to contract by cannot be fixed in advance. Biconnected components are the natural first choice, being structural rather than optimised — a module is a set of proteins that remains connected when any one of them is removed — and on a sparsely studied organism they carry most of the answer, since the reported network is largely bridges (§7). But a well-studied core *is* biconnected: the decomposition returns it unchanged, and a drill-down built on it descends forever into the same picture. We therefore contract by biconnected components when they decompose the graph and by modularity otherwise, using Louvain [9] weighted by trust, so that a community is held together by evidence rather than by edge count. Modularity always divides, and divides again, which is the property the recursion needs.

“Decomposes” has to be stated carefully, and the human interactome shows why. Its biconnected decomposition is not degenerate in the obvious sense — it returns thousands of components, and its largest holds 76% of the proteins rather than all of them — but the remainder are

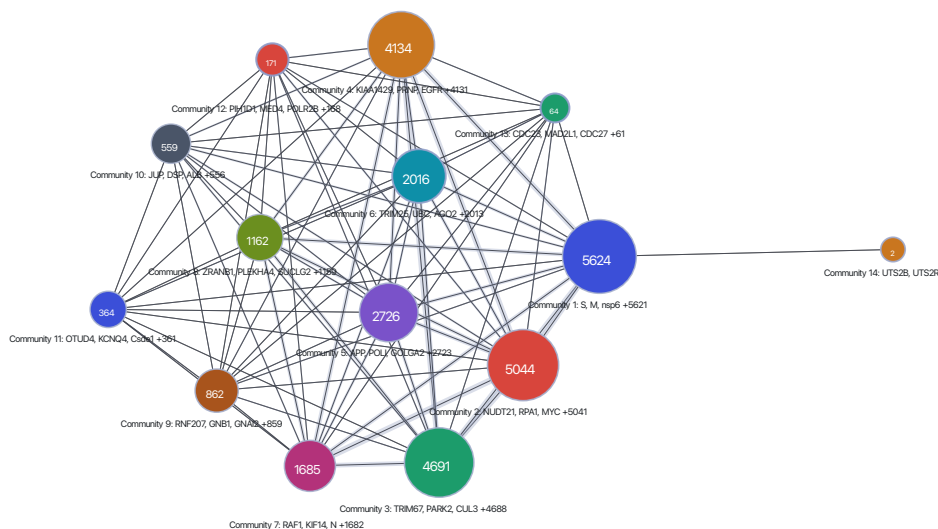


Figure 2: The complete human interactome of BioGRID 5.0.260 — 29,104 proteins and 1,047,820 interactions, no trust threshold — read one level up. Node area is proteins in the module, the number inside is that count, the ring is the mean trust of the module’s internal interactions, and link width is the number of interactions spanning two modules. Fourteen communities, computed in 2.0s. Community 14 is drawn far out because almost nothing connects it, which is a property of the network rather than of the layout.

two-protein bridges, so the drawing is one huge node ringed by specks. We therefore require a *second* module worth drawing: more than one component, none holding more than 90% of the proteins, and a second-largest holding at least 1% of the proteins and at least three of them.

Which grouping this selects is itself a measurement, reported in §7: the structural decomposition is chosen for barely-studied networks, whose reported interactions are little more than a forest of bridges, and modularity for everything that has been studied enough to have a core.

Two details matter for correctness of the picture rather than of the algorithm. Biconnected components share their articulation points, so the groups are made disjoint before drawing — each protein is claimed by the largest group containing it — since a protein drawn in two modules is counted twice, sized twice, and ambiguous to click. And a long tail of two-protein modules is folded into a single node rather than discarded, so no protein leaves the picture as one descends. Our Louvain implementation visits nodes in index order rather than at random, making the partition, and therefore the figure, reproducible.

We add one non-standard variant. Girvan–Newman removal deletes the edge of highest betweenness, identifying joins in the network *as reported*. Removal in ascending order of trust instead deletes the least well-supported edge first, and answers a different question: what structure survives if only the evidence is believed. Both record component count and modularity at each step.

6 Implementation

ProLiVis 2.0 is a static web application: TypeScript, no server, no installation. Data never leaves the user’s machine unless online mode or bibliometric enrichment is enabled, and both are optional.

Bulk BioGRID releases are ingested into DuckDB compiled to WebAssembly, persisted in

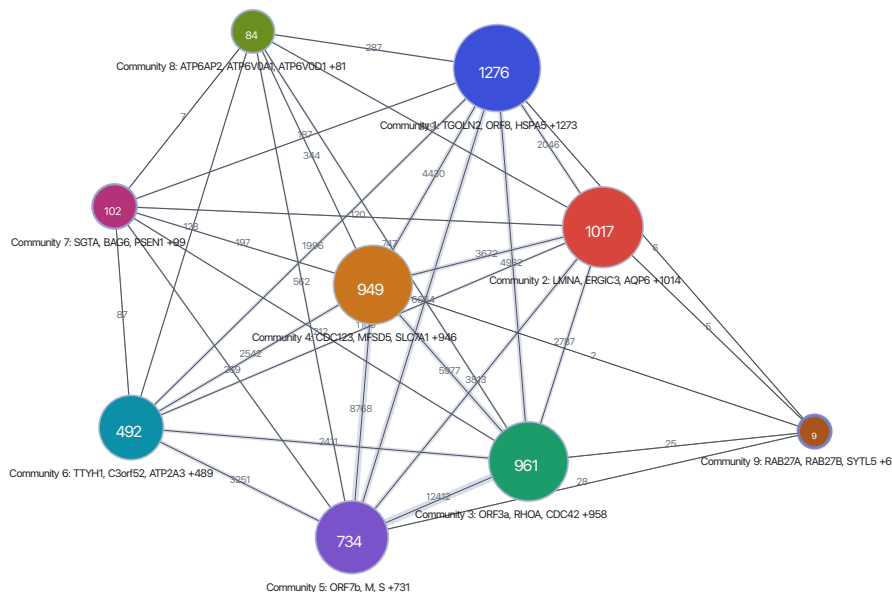


Figure 3: Community 1 of Figure 2 opened: 5,624 proteins and 141,011 interactions, contracted again into nine communities. Link labels are interaction counts. The membership is coherent without having been told anything about biology — a vesicular-transport and ER community (TGOLN2, HSPA5), a v-ATPase community (ATP6AP2, ATP6V0A1, ATP6V0D1), a small RAB27A/RAB27B/SYTL5 secretory community, and two communities carrying the SARS-CoV-2 proteins that this organism's records interact with.

the browser's Origin Private File System. All parsing, coercion and aggregation happen in the database; the application never iterates over interaction records in JavaScript. Archive members are inflated in line-aligned chunks with the reader throttled, so peak memory is bounded by one chunk regardless of archive size.

Columns are resolved by alias onto a canonical set, because BioGRID's tabular formats disagree: the bulk `tab3` files and the `tab2` output of the REST service name six columns differently, and the REST service does not offer `tab3` at all.

Three properties of the data warrant recording, since each is a trap:

- **Publication Source** is not always a PubMed identifier. Approximately a fifth of records in release 5.0.260 are DOI-referenced.
- **Throughput** is a set, not a scalar; a record may be tagged both high and low throughput.
- Gene symbols are not unique across organisms. In the coronavirus release alone, 18 symbols map to more than one gene: **E**, **M**, **N** and **S** exist in three coronavirus species. Nodes are keyed by BioGRID gene identifier; symbol-keying would fuse the SARS-CoV-2 nucleocapsid with the SARS-CoV one.

Every figure carries a session manifest recording release, query, filters, trust configuration and layout options. With a deterministic layout, this is sufficient to regenerate the figure exactly.

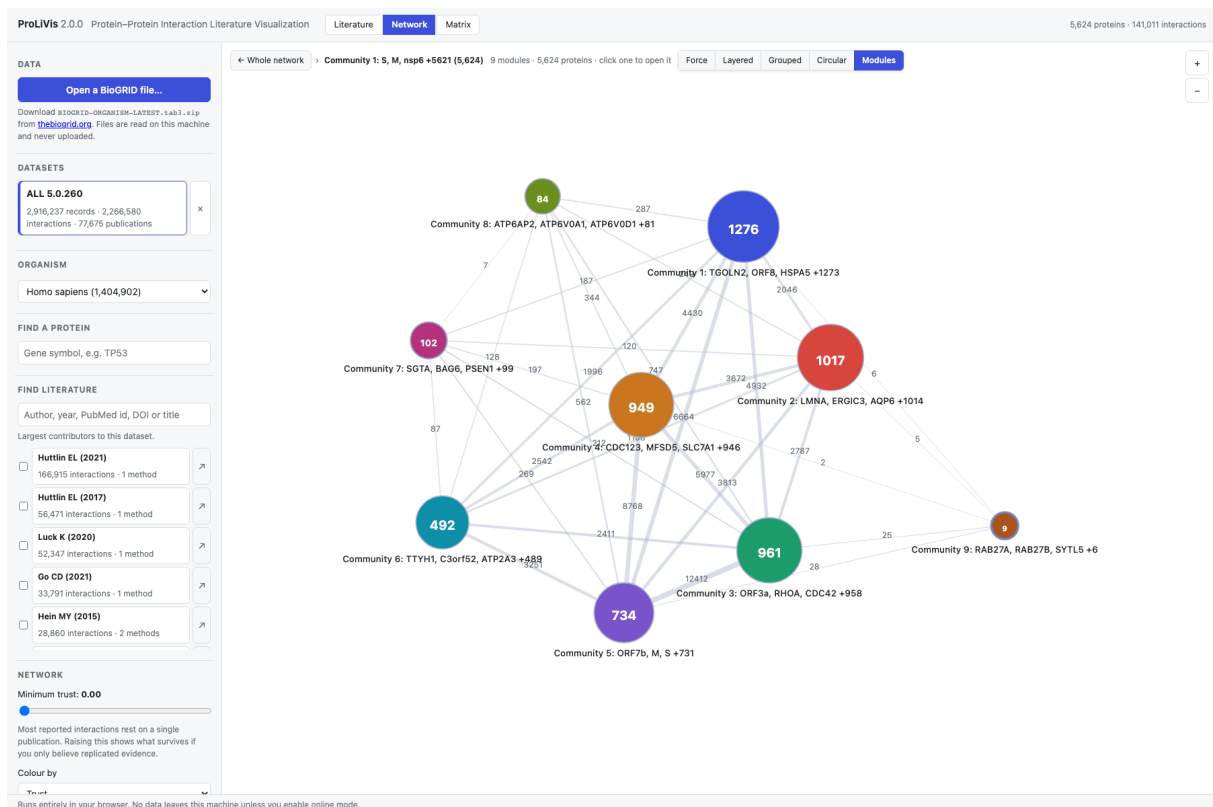


Figure 4: The interface, having opened one community of the human interactome. The breadcrumb above the canvas records the descent and returns to any level of it; the numbers on the links are interaction counts, and the panel on the left is the same trust threshold, density and literature search that apply to every view. The whole application is static files: this is a browser tab, with the release in an embedded analytical database beneath it.

7 Evaluation

Scalability. On a consumer laptop, the complete BioGRID 5.0.260 release (2,916,237 records, 1.55 GB uncompressed) ingests in 78 s, yielding 2,266,580 interactions across 98 organisms. The organism list then returns in 0.7 s and the experimental-system breakdown in 0.4 s; the centre layout for *Homo sapiens* (16,178 nodes) takes 0.8 s. Scoring all 1,068,827 human interactions takes 12 s, after which re-scoring under a changed weight vector is pure arithmetic over gathered evidence — 1.6 ms for 898 interactions — which is what permits interactive re-weighting. Maximal-clique extraction over the human graph thresholded at 0.3 (14,567 proteins, 122,317 interactions) takes 0.4 s.

Reading a large network one level up. Table 1 records the descent through the complete human interactome of release 5.0.260 — 29,104 proteins and 1,047,820 interactions, with no trust threshold applied — opening the largest module at each step, which is the worst case. 4 steps take a reader from a million interactions to a module of 53 proteins that is drawn protein by protein, and the whole descent costs 2.2 s of computation. Figures 2 and 3 are its first two levels.

Which grouping a network needs. Of the 41 organisms in release 5.0.260 whose physical network has at least 100 proteins, 10 decompose into biconnected components under the criterion of §5.1; the rest are contracted by modularity. The split is not arbitrary. The structural ones are the barely-studied networks — *Bos taurus* (605 proteins, 586 interactions), *Danio rerio* (529, 545), *Gallus gallus* (451, 472), Human Herpesvirus 1 (324, 395) — where the reported

Level	Proteins	Interactions	Grouping	Modules	Largest	Time (s)
0	29,104	1,047,820	communities	14	5,624	2.0
1	5,624	141,011	communities	9	1,276	0.2
2	1,276	15,102	communities	10	267	0.0
3	267	1,673	communities	8	53	0.0

After the last step, 53 proteins remain, which is drawn protein by protein.

Table 1: Contracting the human interactome, opening the largest module each time. Generated by the figure pipeline rather than transcribed; the underlying measurements are in `paper/figures/drilldown.json`.

network has scarcely more interactions than proteins and is therefore close to a forest of bridges. The well-studied ones — human, yeast, mouse, fly, *E. coli*, *Arabidopsis* — have a core that is biconnected by construction, and only modularity divides them. A tool that offered biconnected components alone would work on the organisms nobody needs help reading.

Evidence structure. Restricted to SARS-CoV-2 in release 5.0.260, the physical network has 8,510 proteins and 34,540 interactions in 1 connected component, of which 2,848 interactions are bridges and 32 proteins are articulation points. Raising the trust threshold to 0.2 leaves 11,320 interactions among 3,864 proteins — roughly a third. The reported network is a well-connected core wrapped in a large periphery of claims that no second publication has repeated.

Structural recovery. Maximal-clique extraction over the same network finds 19,136 cliques of four or more proteins, the largest having 9 members: GNB2L1, N, ORF6, nsp1, nsp10, nsp12, nsp14, nsp3, nsp5. Without being told any biology, the largest cliques are the SARS-CoV-2 replication–transcription complex — nsp5, nsp9, nsp10, nsp12, nsp13, nsp14, nsp15, nsp16 — together with the host proteins that co-purify with it. The caveat belongs with the result: these interactions come predominantly from affinity-capture screens, in which a clique is evidence of co-purification and not of simultaneous physical contact. The extraction reports what the literature asserts, which is the same discipline the trust model observes.

Preset behaviour. Over all 34,540 interactions, `structural-strict` correlates with the assay-directness term at $r = 0.512$, against `evidence-only`’s $r = 0.302$ and `literature-aware`’s $r = 0.325$. The presets therefore order interactions differently rather than cosmetically, and in the direction their names claim; none of them is a proxy for directness alone.

Discrimination. *To be completed.* The model should be evaluated as a ranker against an external reference set it never saw — CORUM co-complex pairs [7] and hu.MAP [8] are the natural choices — reporting AUROC and average precision per preset and per organism, alongside MIscore [5] as a baseline, and a leave-one-term-out ablation. The harness for this is implemented and tested; the numbers require running it against a reference set and are not reported here. We state this rather than substituting an internally derived reference set, which would be circular.

8 Limitations

The model scores evidence, not biology: a well-replicated artefact scores highly, and should, because the problem lies in the literature and the score reports it faithfully. It ranks interactions BioGRID already lists and never proposes new ones. It cannot see co-citation bias; where a field converges on a belief, the bibliometric terms will agree with the field. Institutional clustering is a proxy for laboratory identity, not a measurement of it: two unrelated groups at one large

university are merged, and a group that has moved institutions is split. The default weights are reasoned, not fitted, and §7 is incomplete until they are tested against an external standard.

9 Availability

Source, documentation and a hosted build are at <https://github.com/melihsozdinler/CenterLayout> under the MIT licence. The archived 1.0 implementation remains in the same repository under its original GPL-3 terms. All figures in this paper are generated by the tool from public BioGRID releases by a single command.

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